

Design and Analysis of a Ladder-Shaped Textile Antenna for SAR-Compliant Driver Fatigue Detection

IEEE Publication | RMKMATE 2025 | Published in IEEE Xplore

Objective

Driver drowsiness is a leading cause of road accidents, and existing detection methods (camera-based eye-tracking, physiological sensors) suffer from high computational cost, privacy concerns, and user discomfort. This work designs a non-intrusive, textile-based passive RF sensor — a ladder-shaped slot antenna integrated into a car seat — that detects driver fatigue through body-coupling effects, monitoring shifts in the S11 reflection coefficient as the driver's posture changes with increasing drowsiness.

Antenna Design

- Ladder-slot patch geometry with 5 impedance-matching slots — engineered to maximize effective radiating surface area within a fixed 5 GHz footprint without increasing physical antenna size
- Dacron substrate selected for $\epsilon_r = 3$, density 1380 kg/m³ — balances low signal distortion with mechanical flexibility required for seamless textile/seat integration
- Annealed copper patch and ground plane for low-loss conductivity at 5 GHz
- Compact form factor: 56×40 mm ground plane, 55×32.86 mm patch — engineered for automotive seat-lining constraints, not just RF performance in isolation

Results

Free-space performance (no human intervention)

- Achieved 5.063 dBi gain and 1.229 VSWR at 5 GHz — VSWR near ideal (1.0) confirms strong impedance match with minimal reflected power
- -19.748 dB reflection coefficient (S11) — indicates efficient power transfer into the antenna, low mismatch loss
- 81.6% total efficiency, 83.2% radiation efficiency — confirms minimal dielectric/conductor loss at the design frequency

On-body performance (simulated on multi-layer human tissue phantom — skin, fat, muscle)

- Body loading shifted resonance frequency and increased return loss, as expected from dielectric loading by human tissue
- Antenna retained high gain and bandwidth despite tissue-induced resonance shift — confirms the design holds up under real dielectric loading conditions, not just idealized free-space simulation

SAR compliance (IEEE/IEC 62704-1, input power constrained to 0.025 W)

- Total SAR: 0.0275707 W/kg; Max 10g SAR: 1.38851 W/kg — both well within FCC safety limits
- Power budget engineered at 0.0224577 W accepted / 0.0181313 W absorbed — demonstrates the design meets safety compliance without over-constraining usable signal power

System Setup / Working

The antenna is embedded in the headrest/backrest of the car seat (upholstered in Dacron for seamless textile integration) — the headrest position is chosen since head movement is the most reliable fatigue indicator. A Vector Network Analyzer (VNA) monitors real-time S11 reflection coefficient variations. The antenna connects to an embedded system via a UFL/SMA connector; the embedded system detects significant S11 shifts indicative of drowsiness and triggers an alert (buzzer, strobe light) to the driver.

My Contribution

- Identified Dacron as the suitable substrate material through literature review of prior research on wearable/automotive textile antennas
- Carried out free-space simulation using a trial-and-error tuning approach to arrive at an optimal antenna configuration
- Performed S11 parameter matching — iteratively adjusting slot dimensions to achieve strong impedance match at the 5 GHz operating frequency
- Developed the body-tissue interaction model — assessing how human tissue loading affects antenna performance — and carried out the SAR compliance analysis

Challenges

- Antenna size at 5 GHz was too small to reliably detect posture-driven fatigue signals — solved by increasing effective surface area through a ladder-slot structure, improving resonance sensitivity without enlarging the physical footprint
- Selecting a substrate with correct dielectric properties for both accurate simulation and real-world implementation — Dacron ($\epsilon_r = 3$) chosen for its low signal distortion, mechanical flexibility, and textile/automotive compatibility
- Maintaining SAR compliance without sacrificing gain — input power was constrained to 0.025 W to keep both total and 10g SAR within FCC limits while preserving usable signal strength
- Modeling realistic on-body loading effects — required building a multi-layer human tissue phantom (skin, fat, muscle) to capture resonance shift and return-loss changes accurately, since free-space simulation alone doesn't reflect real seat-integrated performance

Future Work

- Implementing the antenna as a functioning passive RF sensor in a real vehicle seat, validated with live VNA measurements
- Integrating machine learning on S11 shift patterns to improve fatigue-detection precision beyond fixed-threshold alerting
- Testing performance robustness across varied seat materials, driver body types, and seating postures to validate real-world generalizability